

GE Healthcare

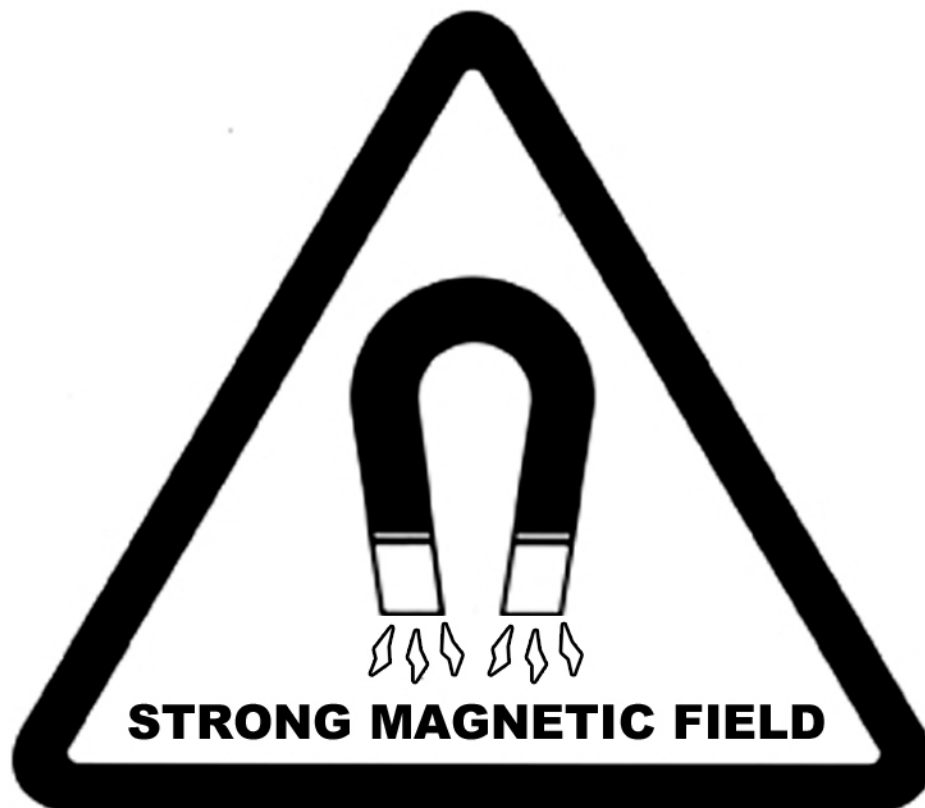
1.5T and 3.0T 6-Channel Flex Coil Service Manual for MR450/MR750/MR450w/GEM



OPERATING DOCUMENTATION

5371824
Revision 2

WARNING



NO PACEMAKERS NO METALLIC IMPLANTS

Persons with pacemakers, neurostimulators or metallic implants must not enter the magnet area. Serious injury may result.



NO LOOSE METAL OBJECTS

Iron and steel materials must not be taken into the magnet area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**ADVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fánleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumų risks apkopes sniedzējam, operatoram vai pacientam.

**ĮSPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi-lačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Rev	Date	Reason for Change
2	May, 2011	Updated for MR450w GEM.
1	July, 2010	Initial release.

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Chapter 1 Introduction

1 1.5T and 3.0T 6-Channel Flex Coil Introduction

The 1.5T and 3.0T 6-Channel Flexible Phased Array Coil is compatible with the GE 1.5T MR450, MR450w, MR450w GEM and the GE 3.0T MR750, MR750w, and MR750w GEM whole body MRI systems. It is a receive only, multi-coil array designed to give the optimal Signal-to-Noise Ratio (SNR) and uniformity for high resolution examinations. The coil enables the imaging of multi-oblique slices of the head and other areas of the body. The multi-element design improves the coverage and resolution.

1.1 Shipping Contents

The table below includes the coil contents shipped for the specified catalog number.

Table 1-1: Shipping Contents by Catalog Number

Description for M0049SS	Part Number	Quantity
1.5T 6-Channel Flex Phased Array Coil with a Quick Disconnect Assembly	5363383	1
Operator Manual	5371823	1
Service Manual	5371824	1
Coil Configuration CD	5372128	1
Description for M0050SS	Part Number	Quantity
3.0T 6-Channel Flex Phased Array Coil with a Quick Disconnect Assembly	5363393	1
Operator Manual	5371823	1
Service Manual	5371824	1
Coil Configuration CD	5372128	1

1.2 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have the notation, "Damage in Shipment," written on all copies of the freight or express bill before delivery is accepted or signed for by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **must** be reported to the carrier immediately upon discovery or, in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and select Option 6.
- Contact your local Service Coordinator for more information on this process.

1.3 Product Locator



NOTICE

The 6-Channel Flex Coil catalog contains three different, serialized components. Make sure to submit three Installation Cards for each catalog.

The Global Install Base Database (also known as the Product Locator System) tracks the shipment, trans-shipment, and field location of the serialized models. There are two methods for submitting Product Locator information.

Method 1: For U.S. ONLY at this time, the preferred method of submitting information is the FE Site Verification website at: <http://gein2.med.ge.com/gib>.

The FE Site Verification has three components available on the web site from the main menu:

- Installation/deinstallation product locator model and serial numbers
- Ability to add/modify the ship to address information
- Ability to obtain a copy of the FE training tutorial for using this web site. A downloadable copy is available on the Product Locator Support Central webpage: <http://supportcentral.ge.com/15563>.

Method 2: Once the Shipping Card is completed and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the Installation Card and extra shipment cards are attached:

- Verify that the serial and model number on each rating plate matches the installation card numbers before removing the installation card.
- Process the Installation Card and discard any extra shipment cards and labels. (There may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination.)

NOTE: There could be as many as three product locator cards, one for each coil half and one for the Quick Disconnect (QD) assembly. Make sure each applicable card is completed and sent in for proper warranty coverage and that the site history is maintained.

1.4 Requirements and Specifications

Coil installation and configuration requires the services of a GE Service Representative. Store the coil in an air-conditioned Scan or Equipment Room.

Table 1-2: Environmental Requirements

Measurement Type	Transport and Storage	Operating Conditions
Barometric pressure	500 to 1060 hPa	700 to 1060 hPa
Humidity	10 to 100% noncondensing	10 to 90% noncondensing

Temperature	-40 to 65° C (-40 to 149° F)	15 to 21° C (59 to 69.8° F)
-------------	------------------------------	-----------------------------

Table 1-3: Coil Frequency Specifications

Field Strength	Coil Model	Frequency
1.5T	6-Channel Flex Phased Array Coil	63.86 MHz $\pm$ 50 kHz
3.0T	6-Channel Flex Phased Array Coil	127.73 MHz $\pm$ 50 kHz

1.5 Coil Description

The 1.5T and 3.0T 6-Channel Flexible Phased Array Coil (shown below) is a receive only, multi-coil array optimized for high resolution exams. The coil enables the user to image multi-oblique slices where the intended areas of interest to be included, but not limited to: head, knee, shoulder, foot, wrist, and other extremities.

Illustration 1-1: 6-Channel Flexible Phased Array Coil - Identical Halves

The 6-Channel Flexible Phased Array Coil consists of two flexible halves that enable large openings and flexibility. The flexible halves conform to the contours of the patient and accommodate the patient's anatomy. This maximizes signal integrity and clinical utility. Both coil halves are connected to the system through an intermediate adapter connector, called a Quick Disconnect (QD) assembly shown below.

NOTE: Coil setup flexibility allows the use of only one coil half if desired.

Illustration 1-2: Quick Disconnect Assembly



NOTE: The Quick Disconnect assembly connects directly to the Low Profile Carriage Assembly (LPCA) of the MRI system shown below.

Illustration 1-3: Low Profile Carriage Assembly



1.6 Modes of Operation

The 6-channel Flexible Phased Array Coil can be operated in multiple modes. Coil modes for neuro applications:

- **6ch Neuro Flex:** Both coil paddles are connected to the QD assembly, and all 6 channels are operational.

- **6ch Neuro Flex1:** Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 1 on QD assembly and three channels are operational.
- **6ch Neuro Flex2:** Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 2 on QD assembly and three channels are operational.
- **3ch Neuro Flex1:** Single paddle connected to Port 1 on QD assembly, and three channels are operational.
- **3ch Neuro Flex2:** Single paddle connected to Port 2 on QD assembly, and three channels are operational.

Coil modes for general purpose applications:

- **6ch GP Flex:** Both coil paddles are connected to the QD assembly, and all 6 channels are operational.
- **6ch GP Flex1:** Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 1 on QD assembly and three channels are operational.
- **6ch GP Flex2:** Both coil paddles are connected to the QD assembly, but only the paddle connected to Port 2 on QD assembly and three channels are operational.
- **3ch GP Flex1:** Single paddle connected to Port 1 on QD assembly, and three channels are operational.
- **3ch GP Flex2:** Single paddle connected to Port 2 on QD assembly, and three channels are operational.

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Chapter 2 Installation

1 1.5T and 3.0T 6-Channel Flex Coil Setup for MCQA Test

1.1 Personnel Requirements

Personnel Requirements	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	60 mins	Not Applicable

1.2 Preliminary Requirements

1.2.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Head TLT Sphere for 1.5T (green solution)	1	-	2135650	-
Head TLT Loader for 1.5T	1	-	2135652	-
Head TLT Sphere for 3.0T (pink solution)	1	-	2359877	-
Head TLT Loader for 3.0T	1	-	2360031	-
Curved Adapter Panel Assembly	1	GEM Systems	5395828	Sumi Motherson

1.2.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Patient Security Strap (provided with system)	1	-	-	-

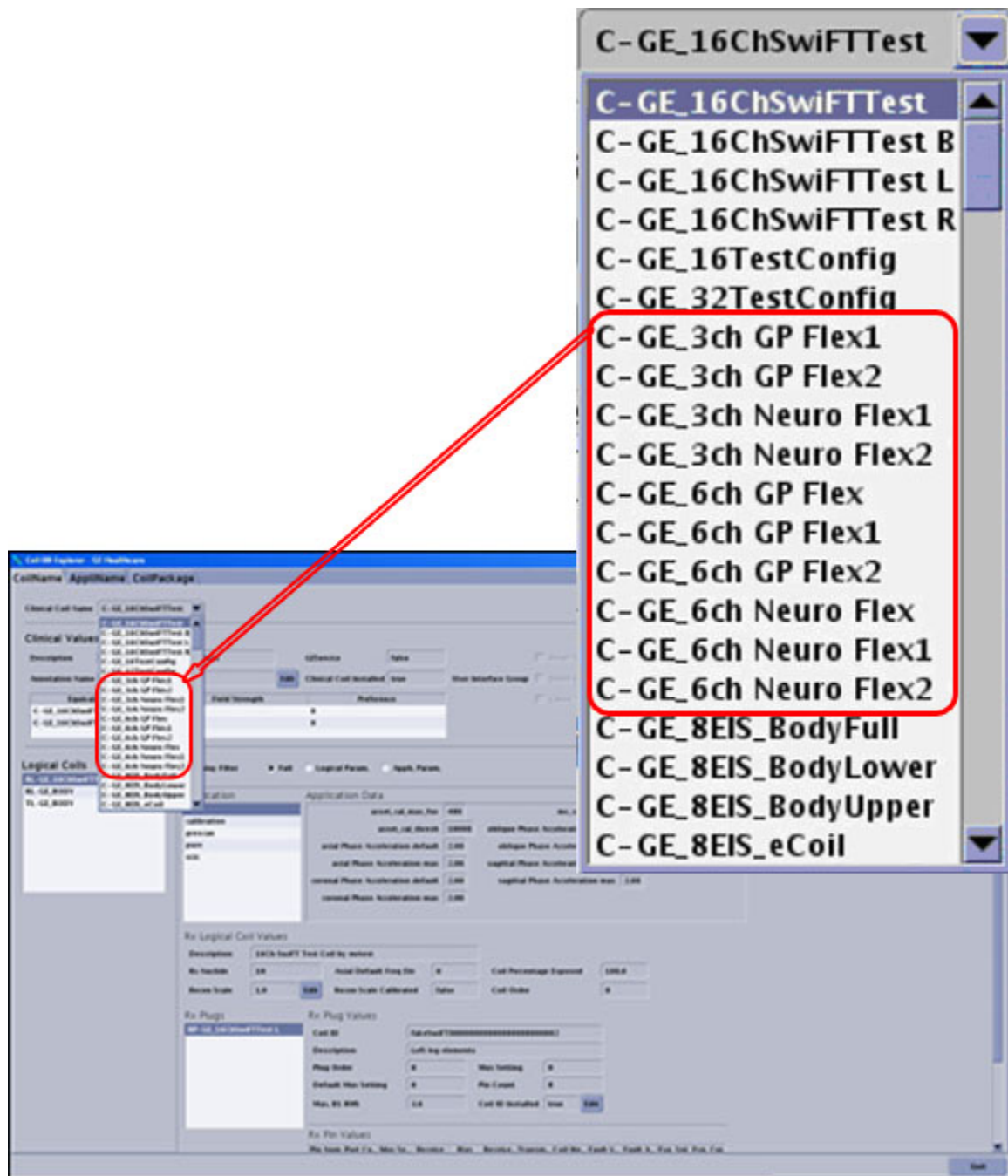
1.3 Procedure

1.3.1 Coil Configuration

Verify the coil configuration for the site. If the applicable 6-Channel Flex Coil is not configured for the site, select the coil from the configuration list:

1. From the Service Desktop, select **Configuration > Coil Database Explorer > Click Here to Start This Tool**.
2. On the *Coil Name* tab, select the pull-down menu for *Clinical Coil Name*.
3. Scroll through the list to verify the existence of the 10 coil modes for the 6-Channel Flex Coils (shown below).

Illustration 2-1: Clinical Coil Names



4. If the coil modes are not in the list, locate the Coil CD that shipped with the coil.
5. To install the 6-Channel Flex Coil configurations and protocols:
 - a. Insert the 6-Channel Flex Coil definitions disk into the default (typically upper) Host DVD drive.
 - b. Open a C-shell from the Service Desktop Manager, and type:

- **su**<Enter>
- <root password> <Enter>
- **mount /mnt/cdrom**<Enter>

NOTE: No action is required when this message displays: Mount : block device/dev/scd0 is write-protected, mounting read only.

- **mnt/cdrom/autorun**<Enter>

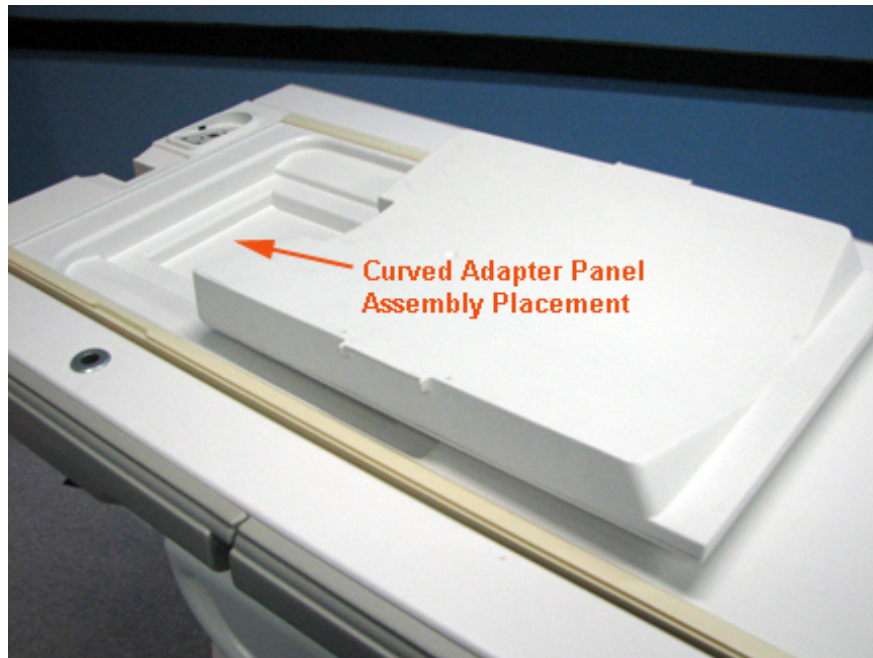
- c. When the *Coil Data Disk Installation* window displays, type **I** to install the CD contents.
- d. When the CD installation is complete, press <Enter> to exit the *Coil Data Disk Installation* window.
- e. Type **eject** into the original C-shell, and remove the CD from the Host DVD drive.
- f. Repeat [Step 1](#) through [Step 3](#) to verify the 6-Channel Flex Coil is in the list.

1.3.2 Flex Coil Alignment

The flex coil must be properly aligned prior to the MCQA test.

1. (For Systems with Flat GEM Table Only) Insert the Curved Panel Assembly into the magnet end of the table as shown.

Illustration 2-2: Curved Adapter Panel Assembly



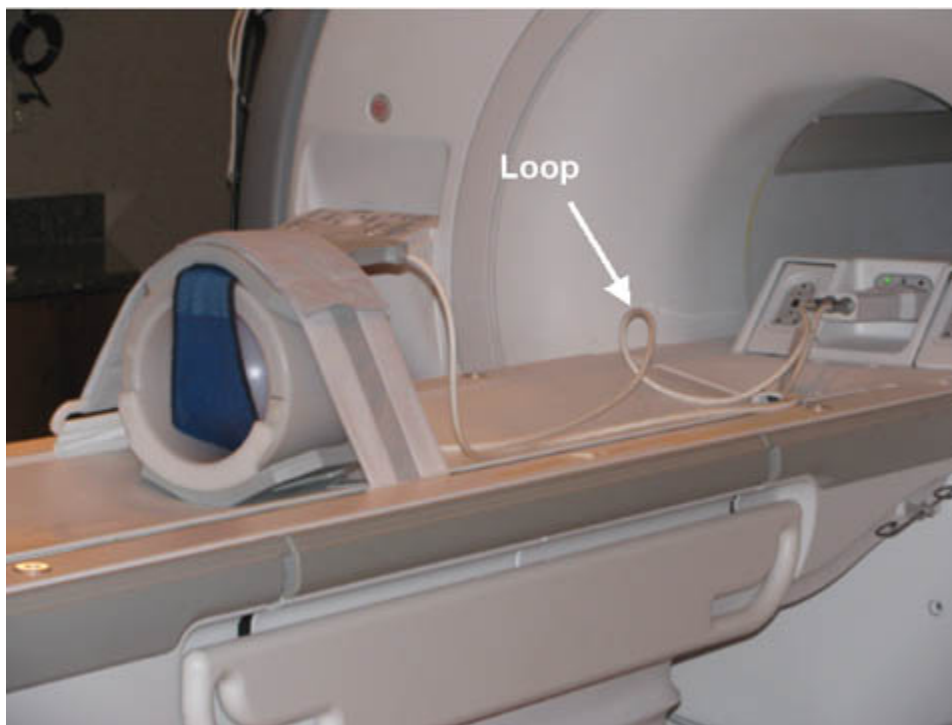
2. Place one coil half onto the table top.

3. Place the phantom with loader onto the coil half.
4. Place the second coil half on top of the phantom loader.

NOTE: The flex coil cables should lay flat on the table top. Make sure these cables do **not** contain loops, such as those shown below.

5. Plug the flex coil Quick Disconnect (QD) into Port A on the Low Profile Carriage Assembly (LPCA).

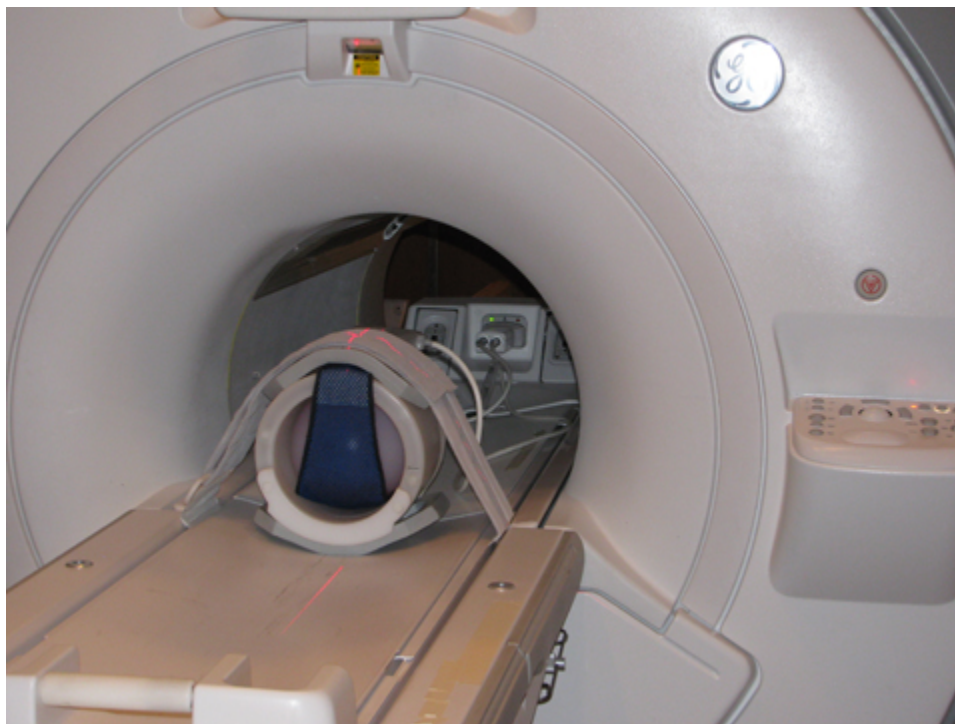
Illustration 2-3: Flex Coil Connected to LPCA



6. Set the landmark using the alignment lights with the crosshairs marked on the anterior half of the coil.
 - a. At the magnet, press the [Alignment Light] button to turn on the laser alignment lights.

NOTE: Make sure the flex coil and phantom with loader are properly aligned (centered left to right) with the laser alignment lights.

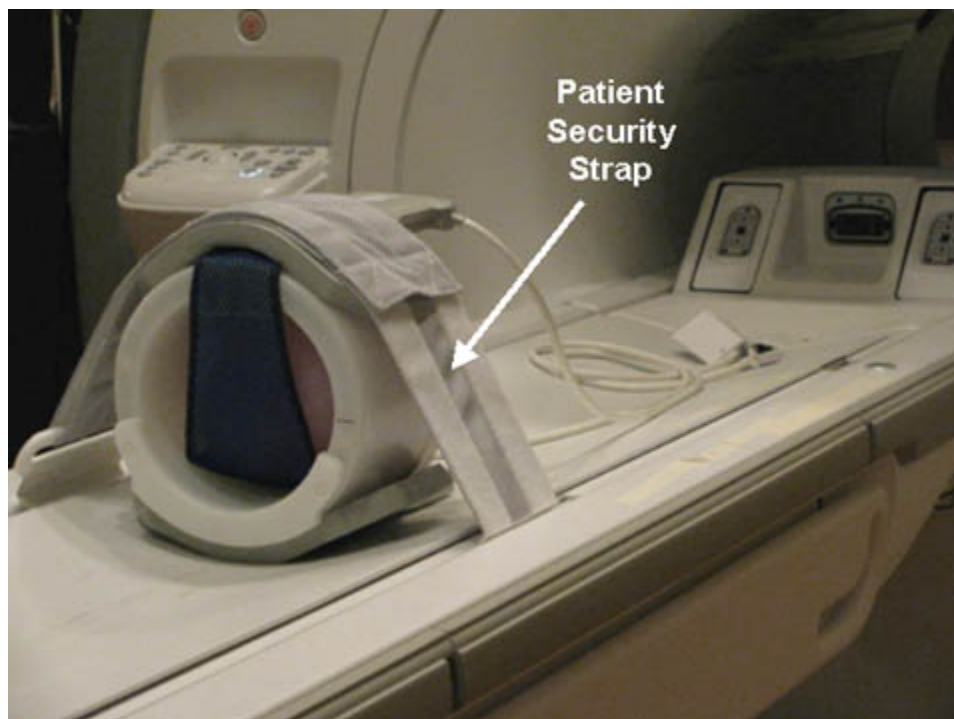
Illustration 2-4: Setting Landmark on Flex Coil



- b. Move the cradle to align the coil landmarks with the laser alignment lights.
 - c. Press the [Landmark] button to save the alignment.
7. Install the patient security strap (provided with the system) to secure both coil halves and the phantom with loader to the table top as shown.

NOTE: Make sure the security strap is properly aligned (centered front to back) on the flex coil and phantom with loader.

Illustration 2-5: Flex Coil and Phantom with Loader



1.3.3 MCQA Test

To perform the Multi-Coil Quality Assurance (MCQA) test for a 1.5T or 3.0T system, locate the appropriate Service Methods DVD:

- Access the **Coil Vendor Manuals** in the Coils folder.
- Navigate to the Flex Coils table.
- Click on the link: *MCQA Tool*.

1.4 Finalization

No finalization steps.

Chapter 3 Replacements

1 1.5T and 3.0T 6-Channel Flex Coil FRUs and Replacement

1.1 Introduction

The two coil halves of the 6-Channel Flexible Phased Array Coil assembly are identical in design and manufacture. The coil halves are not matched pairs, and do not require tuning so individual coil halves can be ordered and replaced.

Each coil half plugs into an Intermediate Adaptor Connector, otherwise known as a Quick Disconnect (QD) assembly, which is then plugged into the Low Profile Carriage Assembly (LPCA) on the scanner. The QD assembly contains six imbedded pre-amps (one for each channel) that can be ordered separately.

Tools Required: 2 Flat-Blade Screwdrivers, 3/16 in. and 1/8 in.

1.2 Safety



NOTICE

To prevent damage to equipment components, observe the following Electro-Static Discharge (ESD) precautions:

- Work on a static-free mat.
- Wear an ESD wrist strap to ensure that any accumulated electro-static charge is discharged from your body to the ground.
- Create a common ground for the equipment to be worked on by connecting the static-free mat, ESD wrist strap and peripheral units to that piece of equipment.

For further information regarding ESD protection, refer to the Safety chapter in this publication.

1.3 Field Replaceable Units

Field Replaceable Units (FRUs) are listed below.

Table 3-1: FRU List

Description	Part Number
1.5T 6-Channel Flex Phased Array coil half (each coil half is identical; therefore, a single half is permitted for replacement)	5229623
1.5T DV 6-Channel Coil QD Assembly	5363382
3.0T 6-Channel Flex Phased Array coil half (Each coil half is identical so a single half may be replaced.)	5229646
3.0T DV 6-Channel Coil QD Assembly	5363382-3
6-Channel Flex Coil Accessories Spares Kit (Needed only when the coil is used on the Surgical Suite product; kit contains replacement parts to support the coil strain relief and mounting mechanism.)	5269901
1.5T Low Noise Pre-Amplifier (Channel location will be marked by the Field Engineer on the pre-amp with a permanent marker at the time of replacement.)	2409869

3.0T Low Noise Pre-Amplifier (Channel location will be marked by the Field Engineer on the pre-amp with a permanent marker at the time of replacement.)	2409871
Head TLT Sphere for 1.5T (Green Solution)	2135650
Head TLT Loader for 1.5T	2135652
Head TLT Sphere for 3.0T (Pink Solution)	2359877
Head TLT Loader for 3.0T	2360031

1.4 Replacement

The 6-Channel Flexible Phased Array Coil assembly consists of two coil halves and the QD assembly. There are no replaceable parts within each of the coil half assemblies.

1.4.1 6-Channel Flex Phased Array Coil Half

The two coil halves of the 6-Channel Flexible Phased Array Coil assembly are identical. Replace one coil half as needed.

1. Make sure the 6-Channel Flexible Phased Array Coil is moved out of the Magnet Room.
2. Disconnect the faulty 6-Channel Flexible Phased Array Coil connector from the QD assembly.
3. Remove the replacement 6-Channel Flexible Phased Array Coil from the shipping container.
4. Connect the new 6-Channel Flexible Phased Array Coil connector to the QD assembly.
5. Place the coil removed from QD assembly into the shipping container.
6. Set up the flex coil and perform the MCQA test as described in [Chapter 2, 1.5T and 3.0T DV 6-Channel Flex Coil Setup for MCQA Test](#).

1.4.2 Quick Disconnect (QD) Assembly

1. Make sure the 6-Channel Flexible Phased Array Coil is moved out of the Magnet Room.
2. Disconnect the 6-Channel Flexible Phased Array Coil first half connector from Port 1 of the QD assembly.

Illustration 3-1: QD Assembly - Port 1 and Port 2



3. Disconnect the 6-Channel Flexible Phased Array Coil second half connector from Port 2 of the QD assembly.
4. Remove the replacement QD assembly from the shipping container.
5. Connect the 6-Channel Flexible Phased Array Coil first half connector to Port 1 of the new QD assembly.
6. Connect the 6-Channel Flexible Phased Array Coil second half connector to Port 2 of the new QD assembly.
7. Place the QD assembly removed from the coil halves into the shipping container.
8. Set up the flex coil and perform the MCQA test as described in [Chapter 2, 1.5T and 3.0T 6-Channel Flex Coil Setup for MCQA Test](#).

1.5 QD Assembly Pre-Amplifiers



NOTICE

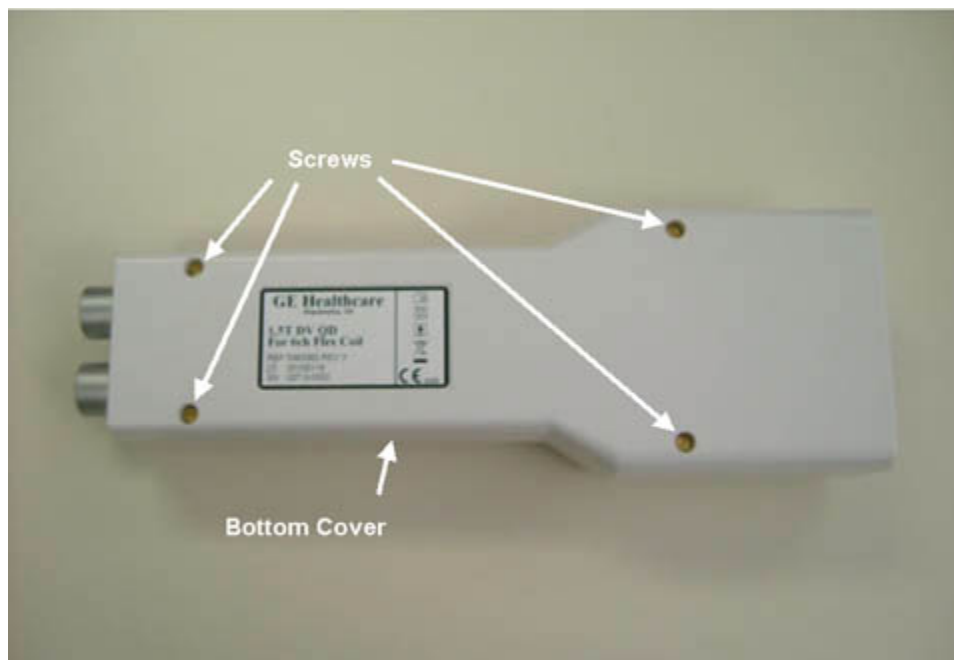
To prevent damage to equipment components, observe the following Electro-Static Discharge (ESD) precautions:

- Work on a static-free mat.
- Wear an ESD wrist strap to ensure that any accumulated electro-static charge is discharged from your body to the ground.
- Create a common ground for the equipment to be worked on by connecting the static-free mat, ESD wrist strap and peripheral units to that piece of equipment.

For further information regarding ESD protection, refer to the Safety chapter in this publication.

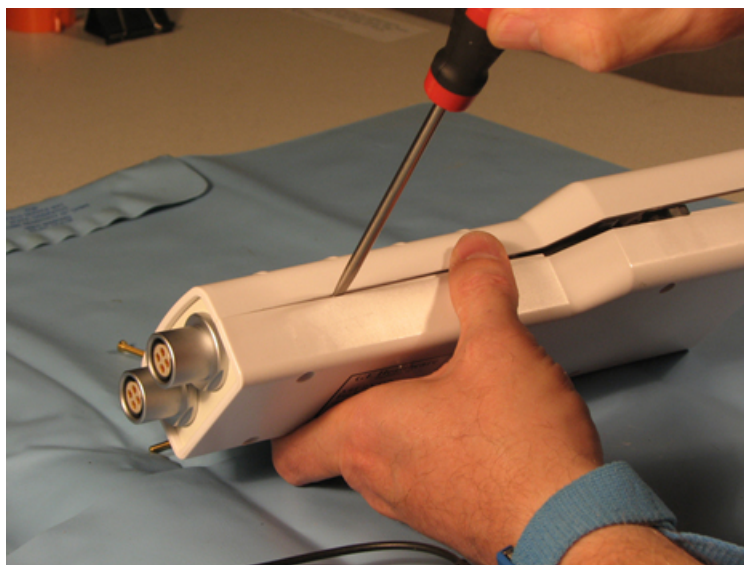
1. Move the QD assembly out of the Magnet Room.
2. Remove the four screws and bottom cover (shown below) from the QD assembly.

Illustration 3-2: QD Assembly - Four Screws and Bottom Cover



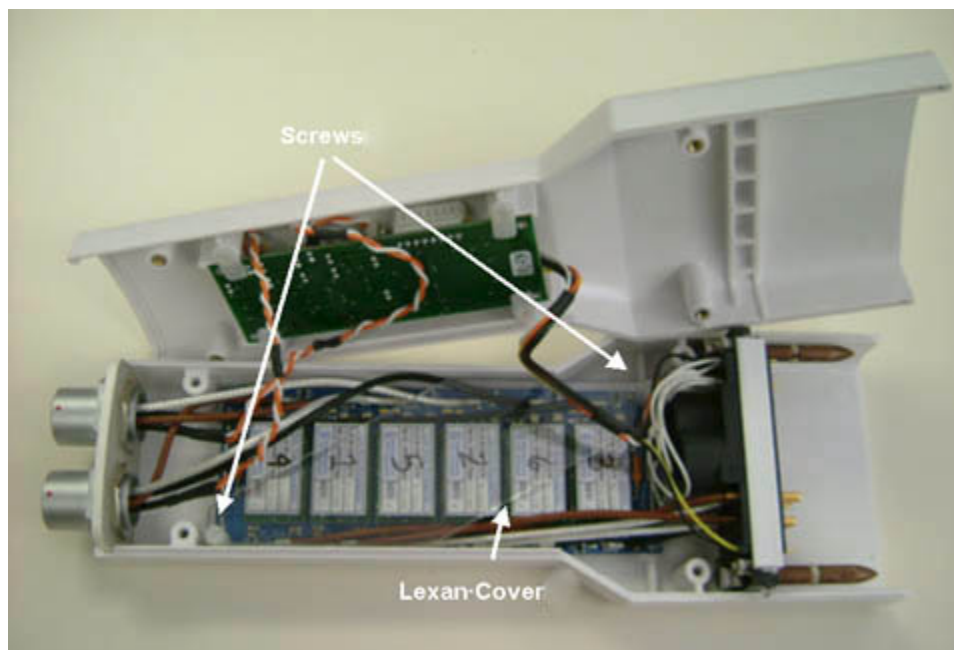
3. If necessary, use a 1/8 in. flat-blade screwdriver and gently pry the housing halves of the QD assembly apart as shown below.

Illustration 3-3: Separating Housing on QD Assembly



4. Remove the two screws and Lexan cover (shown below) from the QD assembly using a 3/16 in. flat-blade screwdriver.

Illustration 3-4: Quick Disconnect (QD) Assembly - Two Screws and Lexan Cover

**NOTICE**

Pay special attention while removing the pre-amp from the socket to avoid damage to the printed circuit, components, or pre-amp.

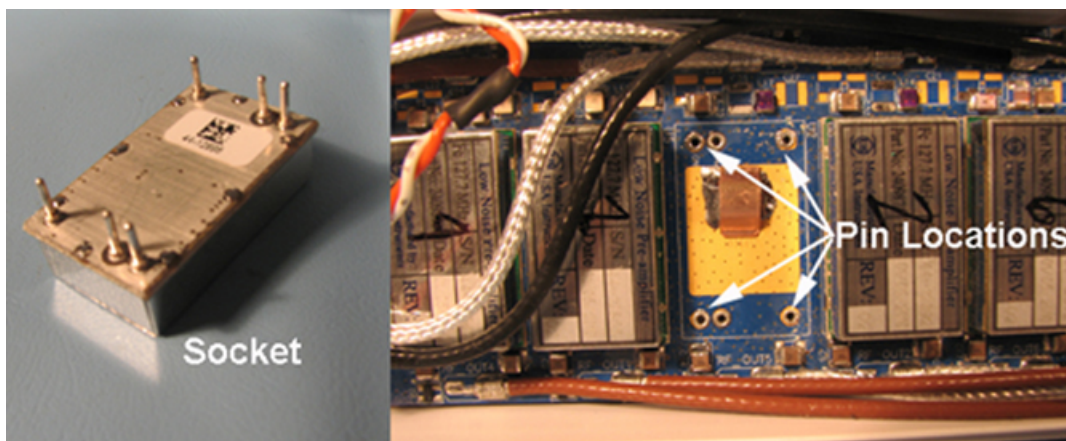
5. Remove the pre-amp from the QD assembly by prying under the pre-amp module on alternate sides of the module with a flat-blade screwdriver as shown.

Illustration 3-5: Removing Pre-Amp from QD Assembly



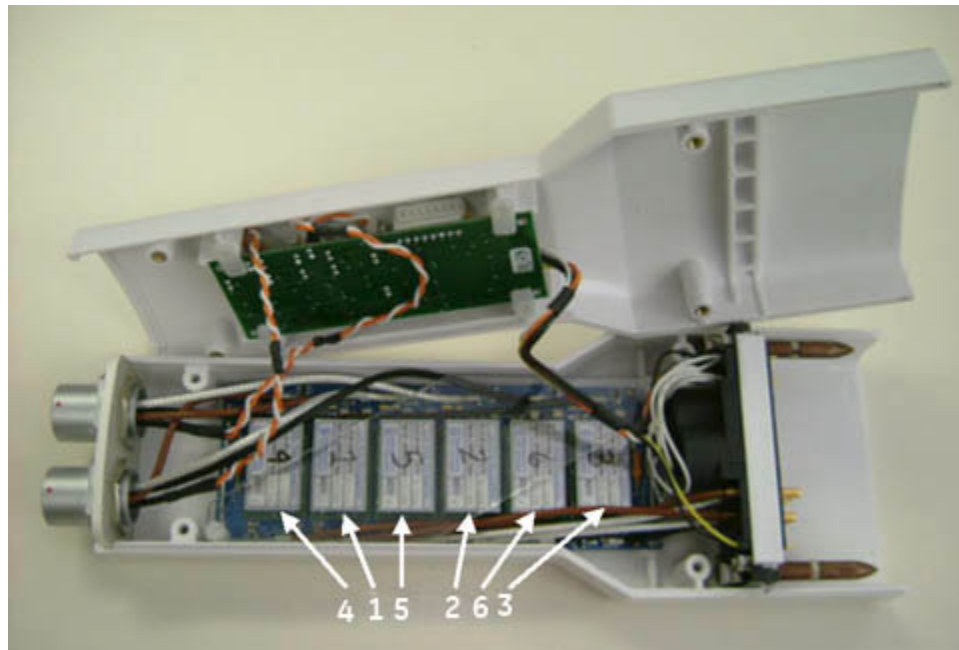
6. Refer to the illustration below to identify the socket and pin locations for the pre-amp module.

Illustration 3-6: Socket and Pin Locations on the Pre-Amp



7. Make a note of the channel (location) number on the pre-amp so the new pre-amp can be marked with the same number.

Illustration 3-7: Quick Disconnect - Preamp Locations for 1.5T and 3.0T



8. Remove the replacement pre-amp from the shipping container and static bag.
9. Install the new pre-amp into the QD assembly, and use a permanent marker to record the channel number on the new pre-amp.

Illustration 3-8: Quick Disconnect (QD) Assembly Pre-Amp - Marking Pre-Amp Number



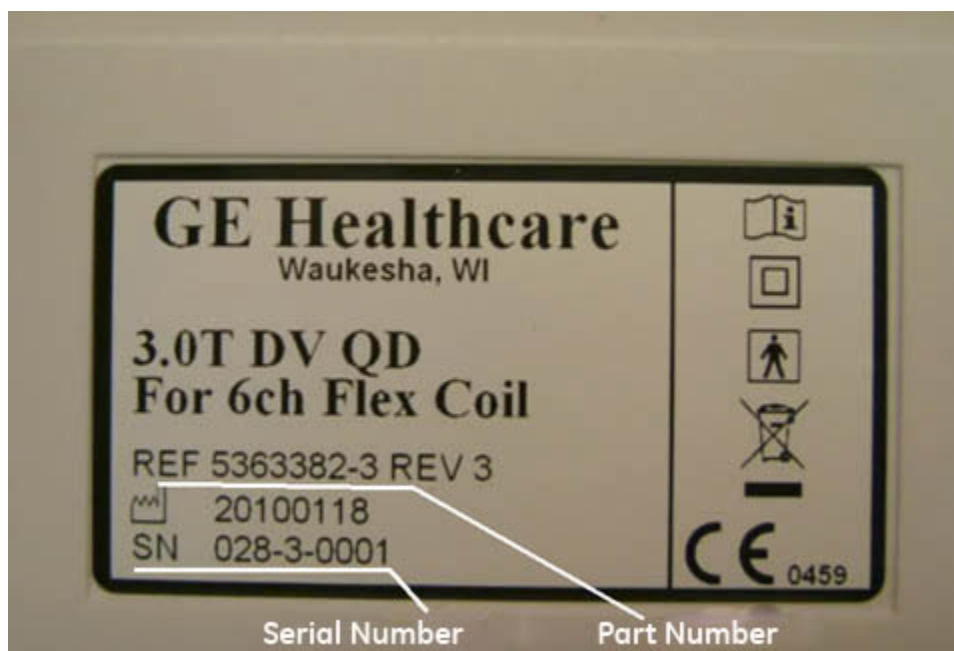
10. Reinstall the Lexan cover and two screws on the QD assembly.
11. Place the pre-amp removed from the QD assembly into the static bag and into the shipping container.

12. Reinstall the bottom cover and four screws to the QD assembly.
13. Set up the flex coil and perform the MCQA test as described in [Chapter 2, 1.5T and 3.0T 6-Channel Flex Coil Setup for MCQA Test](#).

1.6 Finalization

Update the Global Installed Base (GIB) with the new serial number information of the product locator for each new assembly replaced.

Illustration 3-9: Quick Disconnect - ID Label Example



Chapter 4 Troubleshooting

1 1.5T and 3.0T 6-Channel Flex Coil Troubleshooting

The two coil halves of the 6-Channel Flexible Phased Array coil assembly are identical in design and manufacture. The foam sealed housing for the coil provides the required flexibility needed for neuro and general purpose imaging applications. This, however, restricts access to the internal components so troubleshooting the coil is performed at the coil half assembly level. The coil halves are not matched pairs, and do not require tuning so individual coil halves can be ordered and replaced.

To determine if the problem lies in the coil or the receive chain, try swapping the coil cables at the Quick Disconnect (QD) Assembly to see if the problem moves with the cable swap.

- If the problem moves, then the issue is with the coil half.
- If the problem does not move with the cable swap, use the MCRV Tool to determine where in the receive chain the problem lies.

Each coil half plugs into an Intermediate Adaptor Connector, otherwise known as a Quick Disconnect (QD) Assembly, which is then plugged into the Low Profile Carriage Assembly (LPCA) on the scanner. Imbedded pre-amps are provided in the QD Assembly.

1.1 Hardware Inspection

1. Inspect the foam housing of each coil half and look for cracks, discoloration or other damage.
2. Inspect the cables and look for signs of damage such as cuts in the insulation, kinks or twists, especially at the coil and connector entry points.
3. Inspect the cable connector and connector pins. Look for signs of damage to the pins such as corrosion, bent pins, or missing pins.
4. Make sure the field strength labeled on the coil is correct for the scanner it is used on.
5. Inspect the QD Assembly connectors for bent or missing pins.
6. Inspect the QD Assembly housing for cracks, discoloration or other damage.
7. Inspect the connectors on the LPCA and look for damaged or missing pins.
8. Make sure the coil cable connector is fully engaged into the QD Assembly connector.
9. Make sure the QD Assembly is fully engaged into the LPCA connector.

1.2 System Inspection

1. Verify the scanner's operating frequency is within the bandwidth of the coil. Refer to [Chapter 1, 1.5T and 3.0T 6-Channel Flex Coil Introduction](#) for the bandwidth specification for the coil. Corrections to system operating frequency may be necessary to improve coil performance.
2. Review the customer's images and look for shading or artifacts.

1.3 MCQA Analysis Report Overview

The following illustrations depict the type of report resulting from a faulty coil connection.

Illustration 4-1: Loss of Signal on Channel 5

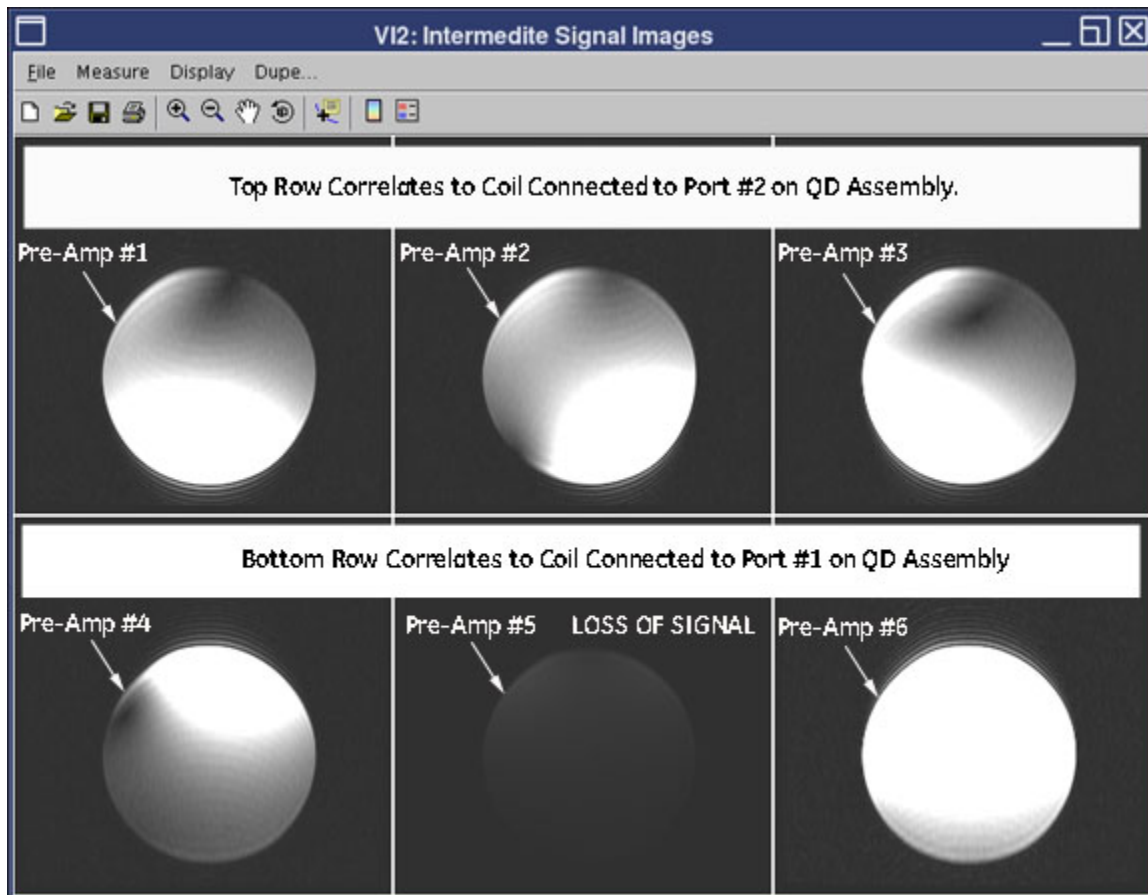


Illustration 4-2: Loss of Signal on Channel 5

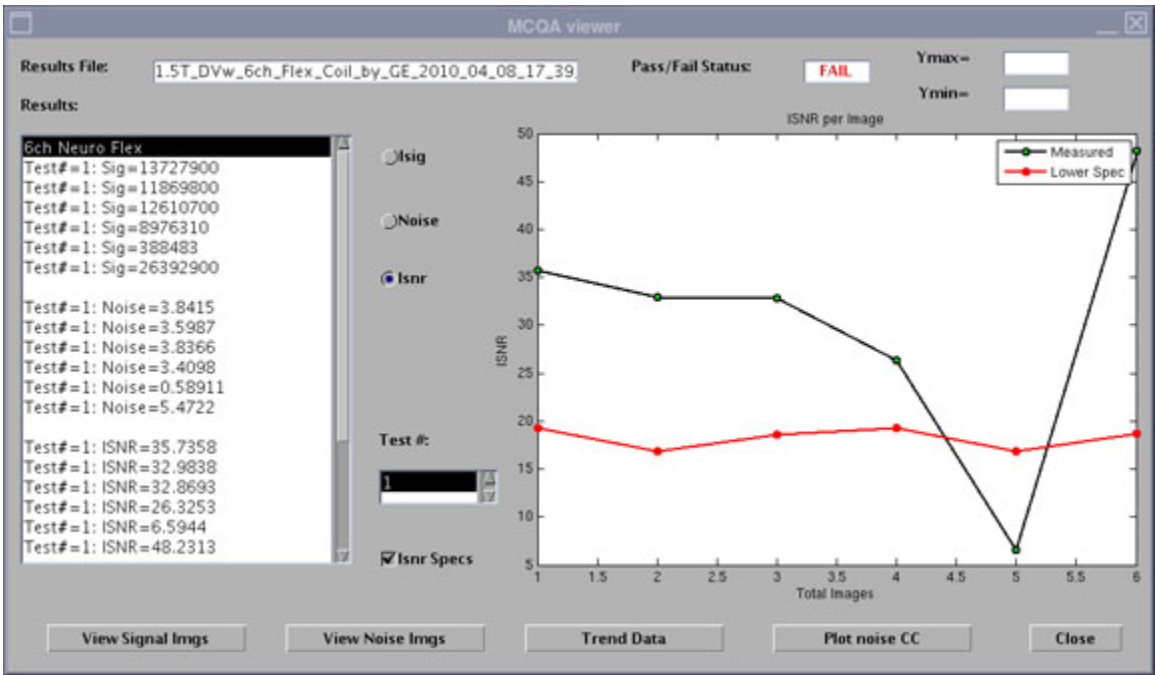


Illustration 4-3: Loss of Signal on Channel 3

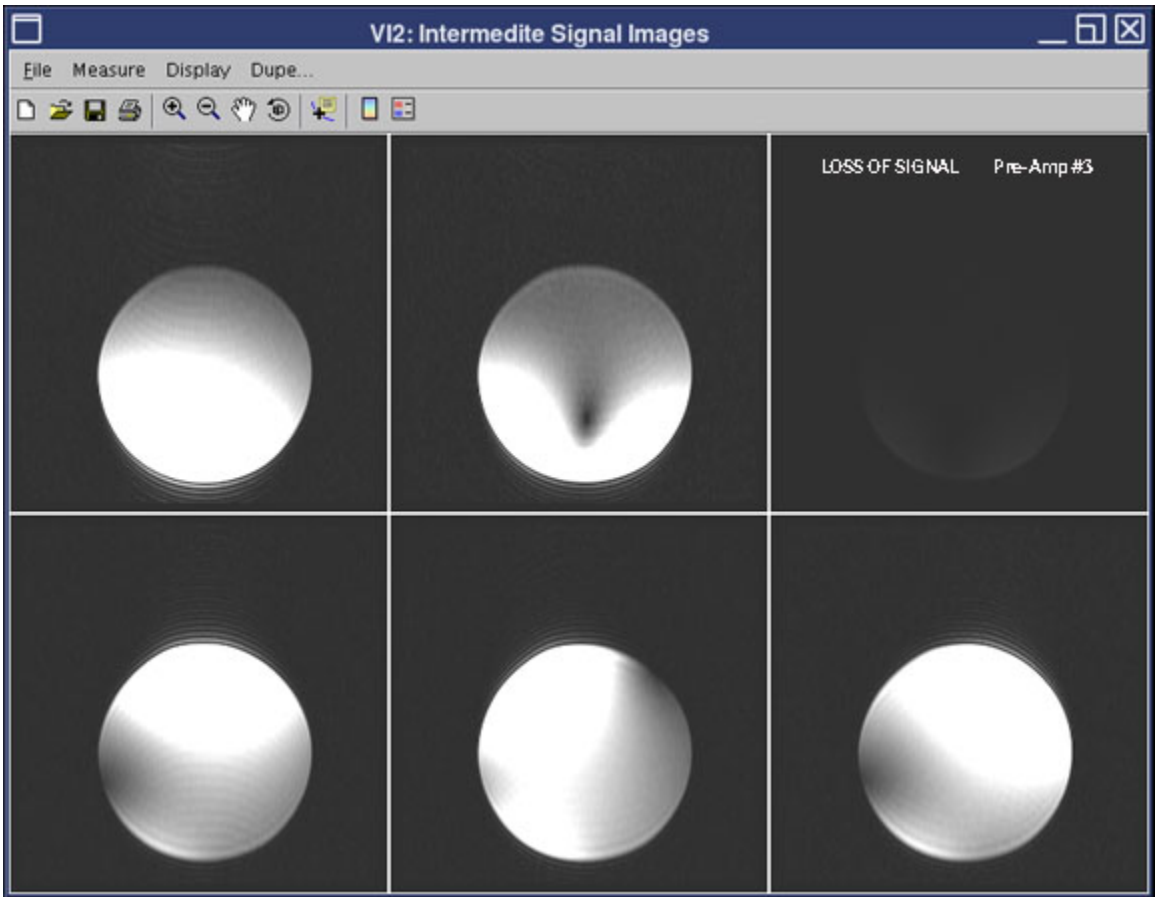
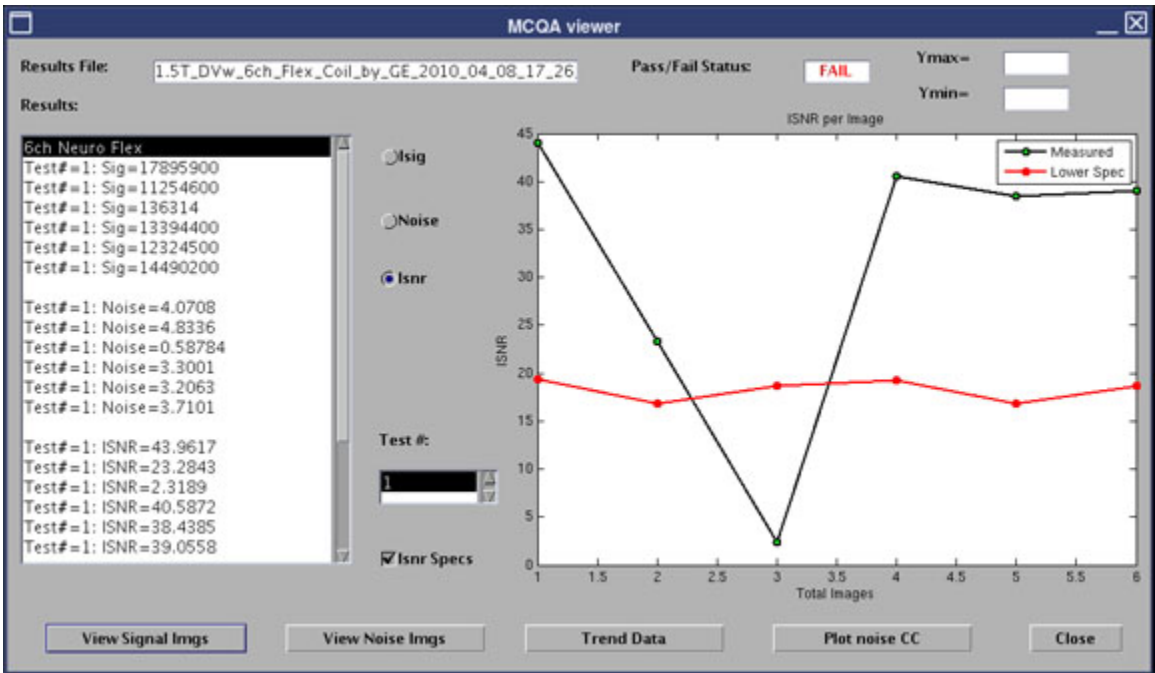


Illustration 4-4: Loss of Signal on Channel 3



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